

ECE 6360 Parallel and Heterogeneous Computing

Homework 4: Diffusion and the Heat Equation

Show that the Euler update step can be written as:

$$T_{i,j}^n = (1 - 4r) T_{i,j}^n + r \left(T_{i+1,j}^n + T_{i-1,j}^n + T_{i,j+1}^n + T_{i,j-1}^n \right)$$

What is r and what value for r satisfies the CFL condition if $\Delta_x = \Delta_y = h$?

What is the largest stable Δ_t for a plate with $\alpha = 1.0 \times 10^{-4} \text{m}^2/\text{s}$ and a spacing of $h = 1\text{mm}$?